

Template Structure + File Roles

THE BIG IDEA

Every file has one job. Know which file to edit before you touch anything.

The BearBots project adds AdvantageKit logging and a clean constants structure to the WPILib template.

KEY CONCEPTS — LESSON 02

Robot.java Lifecycle manager	RobotContainer Wiring hub	Constants.java Source of truth	AdvantageKit Logging layer	Simulator Test bench
Extends LoggedRobot. Runs CommandScheduler. Called every 20ms.	Subsystems created here. Button bindings here. IO layer injected here.	public static final double. k-prefix names the value. One file per subsystem.	Logger.start() at boot. NT4Publisher feeds AScope. Every loop cycle captured.	Simulate before hardware. Drag Keyboard 0 -> Joystick[0]. Graph encoders in AScope.

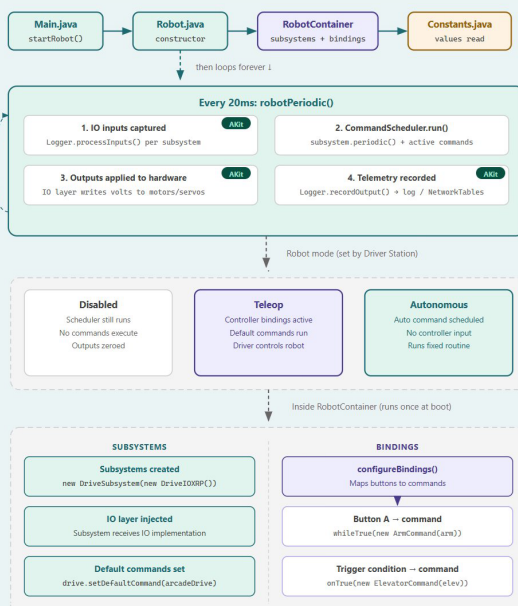
THE SETUP WORKFLOW

- 1 Create XRP-Template project, explore XRP-Template file structure
- 2 Install Git: download, accept defaults, restart VS Code
- 3 git config user.name and user.email — one time per machine
- 4 Clone: git clone <repo url> into C:\FRC
- 5 git pull — get latest files
- 6 Simulate Robot Code, enable halsim_gui
- 7 Connect AdvantageScope, graph LeftPositionMeters
- 8 Drive with WASD, predict graph shapes
- 9 Broken Robot Lab: fix bugs using symptom descriptions only

AFTER THIS LESSON I CAN...

- ☐ Name the job of each file: Main, Robot, RobotContainer, Constants, build.gradle
- ☐ Explain why extends LoggedRobot instead of TimedRobot
- ☐ Use git add / commit / pull to save my work
- ☐ Launch the simulator and connect AdvantageScope
- ☐ Read encoder graphs and describe what the robot is doing
- ☐ Find a bug from a symptom description alone

PROGRAM FLOW



KEY VOCABULARY

LoggedRobot	AdvantageKit base class, wraps each 20ms loop with data capture
RobotContainer	Where subsystems are created and buttons are wired to commands
Constants.java	Fixed values using public static final; k-prefix names; one file per subsystem
git commit	Saves a labeled snapshot of code you can return to
NT4Publisher	Streams AdvantageKit log data to NetworkTables for AdvantageScope

Write anything here — surprises, connections, things to look up later.